








Statistics of Financial Markets

Seminar 8: Volatility estimators



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **BET**: Bucharest Exchange Trading (index)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **DM**: Diebold–Mariano (test)
- **EODHD**: EOD Historical Data (market-data provider)
- **EWMA**: Exponentially Weighted Moving Average
- **GK**: Garman–Klass (volatility estimator)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **LB**: Ljung–Box (test)
- **LM**: Lagrange Multiplier
- **MSE**: Mean Squared Error
- **OHLC**: Open, High, Low, Close
- **OMV**: Österreichische Mineralölverwaltung — Austrian Mineral Oil Administration (OMV group)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **VIX**: CBOE Volatility IndeX
- **YZ**: Yang–Zhang (volatility estimator)

Today's Question and Seminar Structure

- Question of the seminar
  - **how can we measure, from prices, a volatility that we never observe?**
- Outputs to produce
  - the answers to the A exercises
  - the table or figure each B exercise asks for
  - a short argued answer to C1; the errors found in the AI answer of C2 and the corrected answer
  - the interpretation sentences each exercise asks for
- Nothing is handed in and nothing is graded: the seminar is for practice
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and their order: the Exercise Map (next slide)
- Contact: Prof. Daniel Traian Pele, danpele@ase.ro

Exercise Map

Task	Question	Type	Model
A1, A2	historical volatility, annualisation and its standard error	Solved, Proposed	A1
A3, A4	EWMA step by step; half-life	Solved, Proposed	A3
A5, A6	Parkinson, Garman–Klass, Rogers–Satchell and Yang–Zhang from OHLC prices	Solved, Proposed	A5
A7, A8	Ljung–Box on squared returns and the ARCH-LM test	Solved, Proposed	A7
A9, A10	MSE and QLIKE losses of volatility forecasts	Solved, Proposed	A9
B1, B2	rolling and EWMA volatility: S&P 500; BET, Bitcoin	Solved, Proposed	B1
B3, B4	range-based estimators: S&P 500; DAX, Bitcoin, TLV	Solved, Proposed	B3
B5, B6	clustering tests: S&P 500; BET, Bitcoin, TLV, SNP	Solved, Proposed	B5
B7, B8	forecast evaluation with the VIX; the best λ	Solved, Proposed	B7
C1, C2	which estimator for BVB stocks? what is wrong in an AI answer?	Proposed	B3, B7

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the audit of an AI answer

Data Used

Series	Source	Prices used	Period
S&P 500	EODHD	close; OHLC from 2008	1990–2026
DAX	EODHD	close; OHLC from 2006	1990–2026
BET	EODHD	close only	1997–2026
Bitcoin	EODHD	OHLC, 7 days a week	2014–2026
Banca Transilvania (TLV), OMV Petrom (SNP)	EODHD	adjusted OHLC	2010–2026
VIX	EODHD	close	1990–2026

- Daily log returns in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- Stocks: open, high and low multiplied by the same adjustment factor as the close (dividends, splits); TLV without 30–31 May 2016 (Chapter 2)
- In the notebook: `returns('sp500')`, `ohlc('btc')`; no account or key is needed

Prerequisites for Today (1/21): Volatility

- **Volatility**: the standard deviation of the returns, the typical size of a daily move

$$\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{t=1}^{n}(r_t - \bar{r})^2}, \qquad \bar{r} = \frac{1}{n}\sum_{t=1}^{n} r_t$$

  - r_t: the daily log return in % of the Data Used slide, with P_t the closing price of day t
  - n: the number of returns; $\bar{r}$: their mean; the hat marks a value estimated from data
  - the divisor $n-1$ instead of n corrects for the fact that the mean $\bar{r}$ is itself estimated

- **Zero-mean volatility**, usual for daily data

$$\hat{\sigma}_0 = \sqrt{\frac{1}{n}\sum_{t=1}^{n} r_t^2}$$

  - the daily mean (a few hundredths of a %) is negligible next to the daily standard deviation (about 1%)

- **Historical volatility**: the same formula on a **rolling window** of the last n days
  - $n = 21$ (a month), 63 (a quarter) or 252 (a year of trading days); the window moves forward one day at a time

Prerequisites for Today (2/21): Annualisation

- **Square-root-of-time rule**: from a daily to an annual volatility

$$\sigma_{\textbf{year}} = \sigma_{\textbf{day}} \sqrt{A}$$

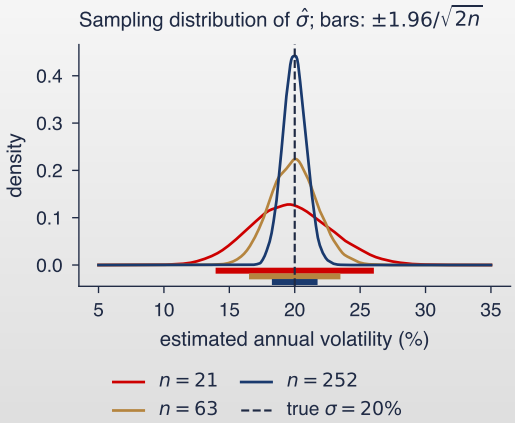
  - A: the number of returns per year; about 252 trading days for stock markets, 365 for Bitcoin (traded 7 days a week)
  - example: 1% per day gives $1 \times \sqrt{252} = 15.9\%$ per year
- Reason: the annual log return is the sum of A daily log returns
  - if the daily returns are uncorrelated and have the same variance, the variances add up:

$$\mathrm{Var}\Big(\sum_{t=1}^{A} r_t\Big) = A\,\sigma^2_{\textbf{day}} \quad \Rightarrow \quad \sigma_{\textbf{year}} = \sqrt{A}\,\sigma_{\textbf{day}}$$

  - with correlated returns the rule is only an approximation
- **The actual frequency** A, counted from the data

$$A = \frac{\text{number of returns in the sample}}{\text{number of years the sample covers}}$$

Prerequisites for Today (3/21): Precision of a Volatility Estimate



- **Standard error** $\mathrm{SE}(\hat{\sigma})$: the standard deviation of $\hat{\sigma}$ over repeated samples
  - for n i.i.d. Normal returns (independent, identically distributed):
$$\frac{\mathrm{SE}(\hat{\sigma})}{\sigma} \approx \frac{1}{\sqrt{2n}}$$
  - $n = 21$: about 15% of σ; $n = 252$: about 4.5%
- **Approximate 95% interval**
$$\hat{\sigma}\left(1 \pm \frac{1.96}{\sqrt{2n}}\right)$$
  - 1.96: the 97.5% quantile of the standard Normal distribution
  - the chart: 20 000 simulated samples with $\sigma = 20\%$; bars: the 95% intervals

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Prerequisites for Today (4/21): EWMA

- **EWMA** (exponentially weighted moving average) variance, RiskMetrics J.P. Morgan/Reuters (1996)

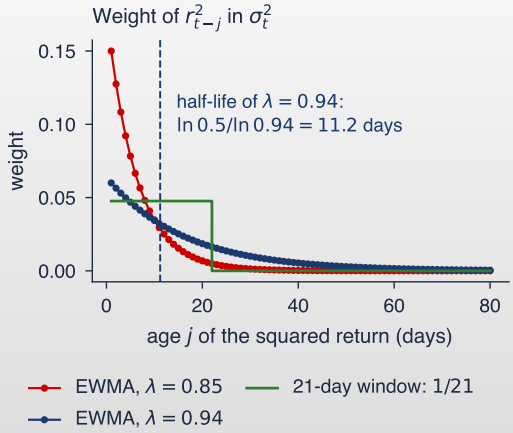
$$\sigma_t^2 = \lambda\, \sigma_{t-1}^2 + (1-\lambda)\, r_{t-1}^2, \qquad 0 < \lambda < 1$$

  - in words: today's variance is a weighted average of yesterday's variance and yesterday's squared return
  - λ: the **decay factor**; $\lambda = 0.94$ for daily data (RiskMetrics)
  - a large λ: a smooth estimate with a long memory; a small λ: a fast reaction to new returns
  - σ_t^2 uses the returns up to day $t-1$ only: it is the forecast of the variance of day t

- Rolling window for comparison: weight $1/n$ for each of the last n days, 0 before

$$\hat{\sigma}_t^2 = \frac{1}{n} \sum_{j=1}^{n} r_{t-j}^2$$

Prerequisites for Today (5/21): EWMA Weights and the Half-Life



- Substituting the recursion again and again:

$$\sigma_t^2 = (1-\lambda)\sum_{j\geq 1} \lambda^{j-1} r_{t-j}^2$$

  - j: the age of the return, in days; its weight $(1-\lambda)\lambda^{j-1}$ falls geometrically; the weights add up to 1

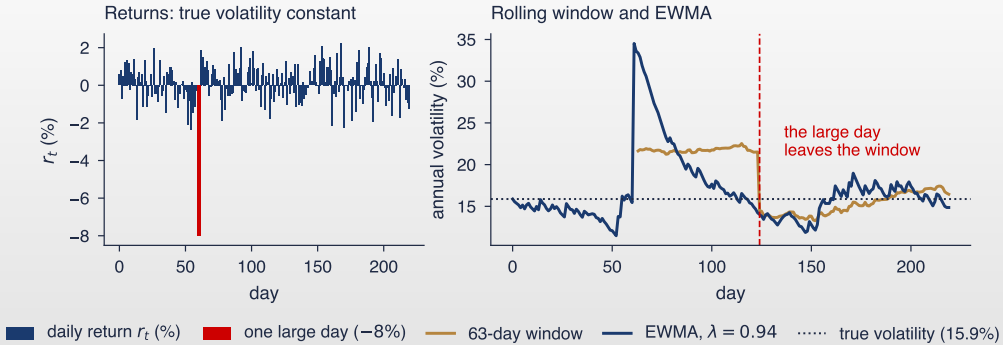
- **Half-life** $h_{1/2}$: the age at which the weight is half of the newest one

$$\lambda^{h_{1/2}} = 0.5 \quad \Rightarrow \quad h_{1/2} = \frac{\ln 0.5}{\ln \lambda}$$

- **Days that carry 99%** of the weight: the last m days carry $1-\lambda^m$

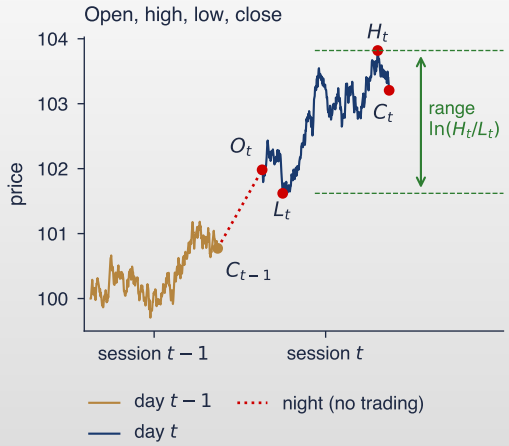
$$1-\lambda^m = 0.99 \quad \Rightarrow \quad m = \frac{\ln 0.01}{\ln \lambda}$$

Prerequisites for Today (6/21): Rolling Window, EWMA and the Ghost Effect



- Simulated returns with a constant volatility of 1% a day and one day of -8%
- **Ghost effect**: the 63-day estimate stays high for exactly 63 days, then drops abruptly when the large day leaves the window; the drop is caused by the calendar, not by news
- EWMA jumps on the next day and then decays smoothly, at the speed set by λ

Prerequisites for Today (7/21): Open, High, Low and Close Prices



- **OHLC** prices of day t: open O_t, high H_t, low L_t, close C_t
  - the session: from O_t to C_t; the night: from C_{t-1} to O_t, without trading
- Log differences in % (all multiplied by 100):
  - **overnight return** $o_t = \ln(O_t/C_{t-1})$
  - $u_t = \ln(H_t/O_t) \geq 0$: the high, measured from the open
  - $d_t = \ln(L_t/O_t) \leq 0$: the low, measured from the open
  - $c_t = \ln(C_t/O_t)$: the open-to-close return of the session
- Two identities
  - close-to-close (**CC**) return: $r_t = \ln(C_t/C_{t-1}) = o_t + c_t$
  - log **range**: $u_t - d_t = \ln(H_t/L_t) \geq 0$

Prerequisites for Today (8/21): Parkinson and Garman–Klass Estimators

- Range-based estimators: each day gives one estimate of the variance; on n days we take the mean
- **Parkinson** Parkinson (1980): the squared range, rescaled

$$\hat{\sigma}_P^2 = \frac{1}{n} \sum_{t=1}^{n} \frac{(u_t - d_t)^2}{4 \ln 2}$$

  - for a price that moves as a Brownian motion without drift, $E[(u_t - d_t)^2] = 4 \ln 2 \, \sigma^2$, so the division by $4 \ln 2 = 2.773$ makes the mean equal to the variance σ^2 of the session
  - the range uses the whole path of the day, not only its two ends: more information than r_t^2
- **Garman–Klass** Garman & Klass (1980): the range plus the open-to-close return

$$\hat{\sigma}_{GK}^2 = \frac{1}{n} \sum_{t=1}^{n} \left[0.5 \, (u_t - d_t)^2 - (2 \ln 2 - 1) \, c_t^2 \right]$$

  - $2 \ln 2 - 1 = 0.386$: the weights 0.5 and 0.386 minimise the variance of the estimator
  - both Parkinson and Garman–Klass assume no drift (no trend within the day)

Prerequisites for Today (9/21): Rogers–Satchell and Yang–Zhang Estimators

- **Rogers–Satchell** Rogers & Satchell (1991): valid also with a drift

$$\hat{\sigma}^2_{RS} = \frac{1}{n}\sum_{t=1}^{n}\Big[u_t\,(u_t - c_t) + d_t\,(d_t - c_t)\Big]$$

  - each term is ≥ 0, because $u_t \geq \max(0, c_t)$ and $d_t \leq \min(0, c_t)$

- **Yang–Zhang** Yang & Zhang (2000) on $n \geq 2$ days: night + session

$$\hat{\sigma}^2_{YZ} = \hat{\sigma}^2_O + k\,\hat{\sigma}^2_C + (1-k)\,\hat{\sigma}^2_{RS}, \qquad k = \frac{0.34}{1.34 + (n+1)/(n-1)}$$

  - $\hat{\sigma}^2_O = \frac{1}{n-1}\sum_t (o_t - \bar{o})^2$, $\hat{\sigma}^2_C = \frac{1}{n-1}\sum_t (c_t - \bar{c})^2$: the sample variances of the night and of the session returns
  - k: the weight that makes the variance of the estimator smallest; for example $k = 0.139$ for $n = 21$
  - Yang–Zhang is the only one of the four estimators that includes the variance of the night

- Annual volatility from any of them: $\sqrt{A\,\hat{\sigma}^2}$

Prerequisites for Today (10/21): Efficiency and Bias of an Estimator

- **Bias**: the systematic error of an estimator

$$\text{bias}(\hat{\sigma}^2) = E(\hat{\sigma}^2) - \sigma^2$$

  - unbiased: on average over many samples, the estimate equals the true variance

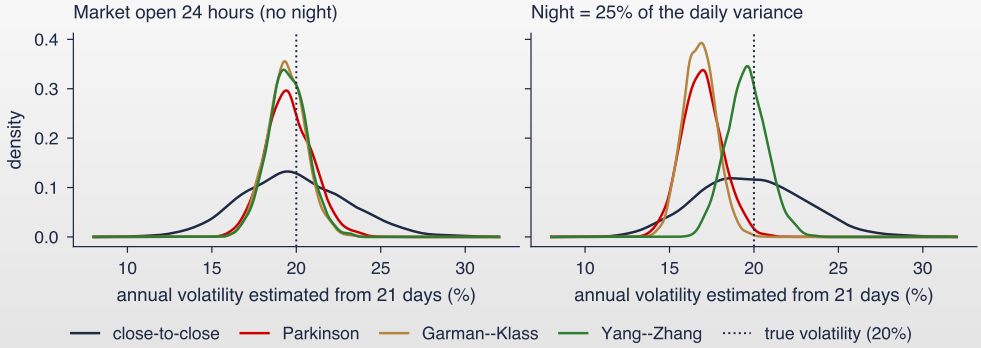
- **Relative efficiency** of an estimator X against close-to-close

$$\text{Eff}(X) = \frac{\text{Var}(\hat{\sigma}^2_{CC})}{\text{Var}(\hat{\sigma}^2_X)}$$

  - $\text{Var}(\hat{\sigma}^2_X)$: how much the estimate varies from one sample to another
  - $\text{Eff}(X) = 5$: X on n days is as precise as close-to-close on $5n$ days
  - for a Brownian motion: Parkinson about 5, Garman–Klass about 7

- Efficiency concerns the **variance** of an estimator; bias concerns its **mean**
  - a precise estimator can still be biased, for example when it ignores part of the daily variance

Prerequisites for Today (11/21): Range-Based Estimators in Simulation



- 3 000 simulated windows of 21 days, true volatility 20%; each curve: the distribution of one estimator
- Left: the range-based estimators are much narrower (more efficient) than close-to-close
- Right: with a night, Parkinson and Garman–Klass measure only the session and are biased downwards; Yang–Zhang stays centred

Prerequisites for Today (12/21): Night and Session in the Close-to-Close Variance

- Since $r_t = o_t + c_t$, the variance of a sum gives

$$\mathrm{Var}(r_t) = \mathrm{Var}(o_t) + \mathrm{Var}(c_t) + 2\,\mathrm{Cov}(o_t, c_t)$$

  - $\mathrm{Var}(o_t)$: the night; $\mathrm{Var}(c_t)$: the session; $\mathrm{Cov}(o_t, c_t)$: how night and session move together
- Sample covariance and correlation

$$\widehat{\mathrm{Cov}}(o, c) = \frac{1}{n-1}\sum_{t=1}^{n}(o_t - \bar{o})(c_t - \bar{c}), \qquad \mathrm{corr}(o, c) = \frac{\widehat{\mathrm{Cov}}(o, c)}{\hat{\sigma}_O\,\hat{\sigma}_C} \in [-1, 1]$$

  - $\mathrm{corr}(o, c) < 0$: the session tends to reverse part of the night move
- **Share of the night**

$$\text{night share} = \frac{\mathrm{Var}(o_t)}{\mathrm{Var}(o_t) + \mathrm{Var}(c_t)}$$

  - a market open 24 hours (Bitcoin) has no night: $o_t \approx 0$

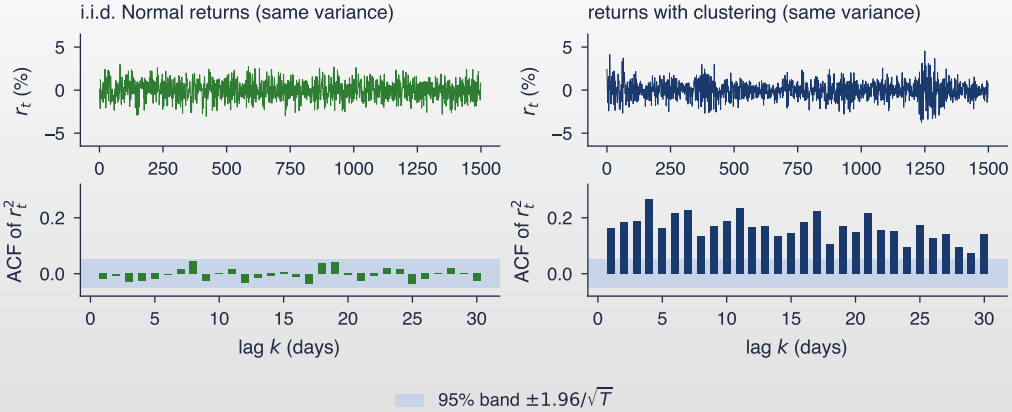
Prerequisites for Today (13/21): Volatility Clustering and the ACF

- **Volatility clustering**: large moves are followed by large moves, small moves by small moves
  - the sign of r_t is almost unpredictable, but its size (r_t^2, $|r_t|$) is predictable
- Sample **ACF** (autocorrelation function) of a series x_t at lag k

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}, \qquad k = 1, 2, \ldots$$

  - x_t: here r_t, r_t^2 or $|r_t|$; T: the number of observations; $\bar{x}$: the mean of the series
  - $\hat{\rho}_k \in [-1, 1]$: the correlation between the series and itself k days earlier
- **95% band** for i.i.d. data: $\pm 1.96/\sqrt{T}$
  - for an i.i.d. series, $\hat{\rho}_k$ is approximately Normal with mean 0 and variance $1/T$; bars outside the band are significant at 5%

Prerequisites for Today (14/21): Volatility Clustering in Simulation



- Same variance, different order of the days: on the right the large days come in groups and the ACF of r_t^2 stays above the band

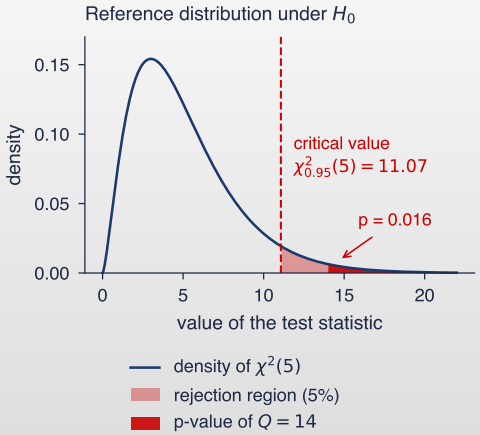
Prerequisites for Today (15/21): The Ljung–Box Test and the χ^2 Distribution

- **Ljung–Box** Ljung & Box (1978): are the first m autocorrelations all zero?

$$Q(m) = T(T+2)\sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{T-k}$$

  - m: the number of lags; $Q(m)$ adds up squared autocorrelations, so it is large when they are large
  - **McLeod–Li** McLeod & Li (1983): the same test on r_t^2; H_0: no volatility clustering

- Under H_0, $Q(m) \sim \chi^2(m)$
  - $\chi^2(m)$: the distribution of a sum of m squared independent standard Normals; its mean is m
  - reject H_0 at 5% if $Q(m) > \chi^2_{0.95}(m)$, the 95% quantile (the critical value)
  - **p-value** $= P(\chi^2(m) \geq Q_{\text{obs}})$: reject at 5% if it is below 0.05



Prerequisites for Today (16/21): The ARCH-LM Test

- **ARCH-LM** Engle (1982): an auxiliary regression of e_t^2 on its own past

$$e_t^2 = a_0 + a_1 e_{t-1}^2 + \cdots + a_q e_{t-q}^2 + u_t, \qquad e_t = r_t - \bar{r}$$

  - q: the number of lags; $a_0, \ldots, a_q$: coefficients estimated by least squares; u_t: the error of the regression

- **Coefficient of determination** R^2

$$R^2 = 1 - \frac{\sum_t \hat{u}_t^2}{\sum_t \left(e_t^2 - \overline{e^2}\right)^2} \in [0, 1]$$

  - $\hat{u}_t$: the residuals; $\overline{e^2}$: the mean of e_t^2; R^2: the share of the variance of e_t^2 explained by its past

- **Test statistic** and decision

$$\mathrm{LM} = n\,R^2 \sim \chi^2(q) \text{ under } H_0 \colon a_1 = \cdots = a_q = 0$$

  - $n = T - q$: the observations used in the regression; H_0: no ARCH effect (constant conditional variance)
  - 5% critical values: $\chi^2_{0.95}(5) = 11.07$, $\chi^2_{0.95}(10) = 18.31$

Prerequisites for Today (17/21): Excess Kurtosis, Shuffling and Weak-Form Efficiency

- **Sample kurtosis**: the mean fourth power of the standardised returns

$$\hat{K} = \frac{1}{T}\sum_{t=1}^{T}\left(\frac{r_t - \bar{r}}{\hat{\sigma}}\right)^4, \qquad \text{excess kurtosis} = \hat{K} - 3$$

  - $K = 3$ for the Normal distribution; an excess > 0: heavy tails, extreme days more frequent than under the Normal
- **Shuffling** (a random permutation) of a series
  - the same T returns, put in a random order; the **seed** fixes the random numbers, so the result can be reproduced
  - a statistic computed before and after shuffling shows whether it depends on the order of the days
- **Weak-form market efficiency** (Chapter 7)
  - past returns cannot be used to predict future returns well enough to earn a profit after costs

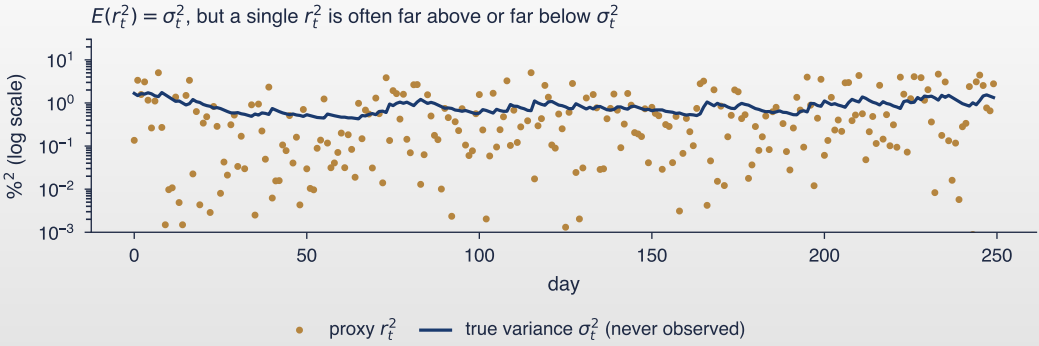
Prerequisites for Today (18/21): Forecast Losses: MSE and QLIKE

- A forecast h_t of the variance of day t, made at the close of day $t-1$, is compared with a **proxy** of the true variance
  - the true variance σ_t^2 is never observed; the usual proxy is r_t^2, with $E(r_t^2 \mid \text{past}) = \sigma_t^2$ when the mean is zero
  - r_t^2 is unbiased but very noisy (next slide)
- Two **loss functions** over N forecast days

$$\text{MSE} = \frac{1}{N}\sum_t \left(r_t^2 - h_t\right)^2, \qquad \text{QLIKE} = \frac{1}{N}\sum_t \left(\frac{r_t^2}{h_t} + \ln h_t\right)$$

  - **MSE** (mean squared error): the mean squared distance between proxy and forecast
  - **QLIKE** Patton (2011): the negative Gaussian log-likelihood, up to constants; its expected value is smallest at $h_t = \sigma_t^2$
  - for both losses, lower is better; both rank forecasts correctly even with the noisy proxy r_t^2

Prerequisites for Today (19/21): The Proxy r_t^2



- Simulated true variance (line) and the proxy r_t^2 (dots), on a log scale
- One day says little: losses must be averaged over many days, and their differences must be tested

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Prerequisites for Today (20/21): The Diebold–Mariano Test

- **DM** (Diebold–Mariano) test Diebold & Mariano (1995): do two forecasts A and B have the same expected loss?

$$d_t = L_t^A - L_t^B, \qquad \bar{d} = \frac{1}{N}\sum_{t=\mathbf{1}}^{N} d_t, \qquad t_{DM} = \frac{\bar{d}}{\widehat{\mathrm{SE}}(\bar{d})}$$

  - L_t^A, L_t^B: the losses of A and B on day t (the MSE or QLIKE terms); d_t: their difference
  - $\widehat{\mathrm{SE}}(\bar{d})$: the standard error of the mean difference, computed so that it remains valid when d_t is autocorrelated and heteroskedastic (HAC)

- Decision
  - $H_\mathbf{0}$: $E(d_t) = 0$; under $H_\mathbf{0}$, t_{DM} is approximately standard Normal
  - $|t_{DM}| > 1.96$: the two forecasts differ at 5%; $t_{DM} < 0$: A has the lower losses
  - forecasts must use only the information available at the close of day $t - 1$

Prerequisites for Today (21/21): The VIX and Realised Volatility

- **VIX** Cboe (2026): the volatility of the S&P 500 expected over the next 30 days, implied by option prices, in annualised %
  - as a forecast of the daily variance (in $\%^2$), with $A = 252$:

$$h_t = \frac{\mathrm{VIX}_{t-1}^2}{A}$$

  - example: $\mathrm{VIX} = 20$ gives $h_t = 400/252 = 1.59$, a daily volatility of 1.26%
- **Realised volatility** of the next 21 days, annualised

$$\mathrm{RV}_t = \sqrt{\frac{A}{21} \sum_{j=1}^{21} r_{t+j}^2}$$

  - **variance risk premium**: the difference between the implied variance (VIX^2) and the variance realised afterwards
- Thinly traded stocks
  - share of days with an unchanged close: $\frac{1}{T} \sum_t 1\{C_t = C_{t-1}\}$, where $1\{\cdot\}$ is 1 if the condition holds and 0 otherwise
  - **stale open**: an opening price equal to the previous close because no trade took place at the open

A1: Historical Volatility from Five Days [Solved]

Task

- Daily returns (%) of a stock: 1.2, −0.8, 0.5, −1.5, 0.6.
- 1. Compute the mean and the sample standard deviation (divide by $n-1$).
- 2. Annualise the standard deviation with $A = 252$.
- 3. Compute the zero-mean volatility of Prerequisites for Today (1/21).
- 4. Compute $1/\sqrt{2n}$ and the approximate 95% interval of Prerequisites for Today (3/21) for the annual volatility.
- Report: the mean, two daily volatilities, one annual volatility, one interval and one sentence.

Solution

- 1. Mean $= 0/5 = 0$; squared deviations 1.44, 0.64, 0.25, 2.25, 0.36, sum 4.94; $s^2 = 4.94/4 = 1.235$, $s = 1.111\%$
- 2. $1.111 \times \sqrt{252} = 17.6\%$
- 3. $\sqrt{4.94/5} = 0.994\%$: with a zero mean, only the divisor changes
- 4. $1/\sqrt{10} = 31.6\%$; $17.6 \times (1 \pm 1.96 \times 0.316)$: $[6.7\%; 28.6\%]$
- Five days say almost nothing: the annual volatility could be anywhere between 6.7% and 28.6%.

Back Exercise list

A2: Bitcoin over Six Days [Proposed]

Solution discussed in class

Task

- Daily Bitcoin returns (%): 3.1, −4.2, 1.0, 5.5, −2.4, −0.9. Model: A1.
- 1. Compute the mean and the sample standard deviation.
- 2. Annualise it with $A = 365$ and, wrongly, with $A = 252$.
- 3. Compute the approximate 95% interval for the annual volatility ($A = 365$).
- Report: two annual volatilities, one interval and one sentence.

A3: EWMA Step by Step [Solved]

Task

- $\lambda = 0.94$; today's EWMA variance is $\sigma_1^2 = 1.00$ ($\%^2$); the next three returns are $r_1 = -2.0\%$, $r_2 = 0.5\%$, $r_3 = 3.0\%$.
- 1. Compute σ_2^2, σ_3^2 and σ_4^2.
- 2. Annualise σ_4 with $A = 252$.
- 3. Compute the half-life and the weight of r_3^2 in σ_4^2.
- Report: three variances, one annual volatility, the half-life and one sentence.

Solution

- 1. $\sigma_2^2 = 0.94 \times 1.00 + 0.06 \times 4.00 = 1.1800$
- $\sigma_3^2 = 0.94 \times 1.1800 + 0.06 \times 0.25 = 1.1242$; $\sigma_4^2 = 0.94 \times 1.1242 + 0.06 \times 9 = 1.5967$
- 2. $\sigma_4 = 1.264\%$; $1.264 \times \sqrt{252} = 20.1\%$
- 3. $\ln 0.5 / \ln 0.94 = 11.2$ days; the newest squared return gets weight $1 - \lambda = 0.06$
- One 3% day raises the variance by 42%; a quiet day lowers it by only 6%.

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A4: A Slower EWMA [Proposed]

Solution discussed in class

Task

- The same data as A3, with $\lambda = 0.97$ (the RiskMetrics monthly value). Model: A3.
- 1. Compute σ_2^2, σ_3^2 and σ_4^2 and annualise σ_4.
- 2. Compute the half-life and the number of days that carry 99% of the weight.
- 3. Compare with A3: which λ reacts faster to the 3% day?
- Report: three variances, one annual volatility, two numbers of days and one sentence.

A5: Four Estimators from One Day [Solved]

Task

- Previous close 100.0; open 101.0, high 103.5, low 99.8, close 102.2.
- 1. Compute o, u, d, c (in %) and $r = o + c$.
- 2. Compute the one-day CC, Parkinson, Garman–Klass and Rogers–Satchell variances.
- 3. Annualise each with $A = 252$, as if every day looked like this one.
- 4. Say which estimators see the overnight move o.
- Report: five log returns, four variances, four annual volatilities and one sentence.

Solution

- 1. $o = 100 \ln(101/100) = 0.995$, $u = 2.445$, $d = -1.195$, $c = 1.181$, $r = 2.176$
- 2. CC $r^2 = 4.736$; Parkinson $3.640^2/2.773 = 4.780$; GK $0.5 \times 3.640^2 - 0.386 \times 1.181^2 = 6.087$; RS 5.931
- 3. $\sqrt{252\,\hat{\sigma}^2}$: CC 34.5%, Parkinson 34.7%, GK 39.2%, RS 38.7%
- 4. Only CC (and Yang–Zhang, on several days) include o; Parkinson, GK and RS see only the session.

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A6: Three Days and the Yang–Zhang Estimator [Proposed]

Solution discussed in class

Task

Day	Open	High	Low	Close
1	50.2	51.0	49.6	50.8
2	50.5	51.9	50.3	51.6
3	51.9	52.4	50.9	51.1

- Previous close: 50.0. Model: A5.
- 1. Compute o, u, d, c for each day.
- 2. Compute the Parkinson and Rogers–Satchell terms of each day and their means.
- 3. Compute k, $\hat{\sigma}_O^2$, $\hat{\sigma}_C^2$ and the Yang–Zhang variance for $n = 3$.
- 4. Compare with the CC sample variance and annualise all three with $A = 252$.
- Report: a small table, three variances, three annual volatilities and one sentence.

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A7: Two Tests for Clustering [Solved]

Task

- $T = 1000$ daily returns; the ACF of the **squared** returns at lags 1–5: 0.20, 0.15, 0.12, 0.10, 0.08; the ARCH-LM regression with $q = 5$ lags on $n = 995$ observations has $R^2 = 0.08$.
- 1. Compute the Ljung–Box statistic $Q(5)$ of the squared returns.
- 2. Compute $\mathrm{LM} = nR^2$.
- 3. Compare both with $\chi^2_{0.95}(5) = 11.07$ and decide.
- Report: two statistics, two decisions and one sentence.

Solution

- 1. Terms $1000 \cdot 1002\, \hat{\rho}_k^2/(1000 - k)$: 40.12, 22.59, 14.47, 10.06, 6.45; $Q(5) = 93.69$
- 2. $\mathrm{LM} = 995 \times 0.08 = 79.60$
- 3. Both far above 11.07 (both $p < 0.001$): reject H_0; the variance is predictable from the recent past.

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A8: Conflicting Results of the Two Tests [Proposed]

Solution discussed in class

Task

- $T = 500$; ACF of the squared returns at lags 1–5: 0.12, 0.08, 0.06, 0.05, 0.04; ARCH-LM with $q = 5$ on $n = 495$ observations: $R^2 = 0.018$. Model: A7.
- 1. Compute $Q(5)$ and decide at 5%.
- 2. Compute LM and decide at 5%.
- 3. Give one reason why the two tests can disagree.
- Report: two statistics, two decisions and one sentence.

A9: MSE and QLIKE by Hand [Solved]

Task

- Over four days the proxy r_t^2 is 1.0, 4.0, 0.25, 2.25 ($\%^2$). Forecast A is flat at 1.5; forecast B is 1.0, 2.5, 0.8, 1.8.
- 1. Compute the MSE of each forecast.
- 2. Compute the QLIKE terms of Prerequisites for Today (18/21) and their mean for each forecast.
- 3. Say which forecast is better under each loss.
- Report: two MSE values, two QLIKE values and one sentence.

Solution

- 1. A: $(1-1.5)^2 = 0.2500$, 6.2500, 1.5625, 0.5625, MSE 2.156; B: 0.0000, 2.2500, 0.3025, 0.2025, MSE 0.689
- 2. A: $1/1.5 + \ln 1.5 = 1.072$, 3.072, 0.572, 1.905, mean 1.655; B: 1.000, 2.516, 0.089, 1.838, mean 1.361
- 3. B is better under both losses: it follows the changes of the variance; a flat forecast misses them.

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A10: Under- and Over-Prediction [Proposed]

Solution discussed in class

Task

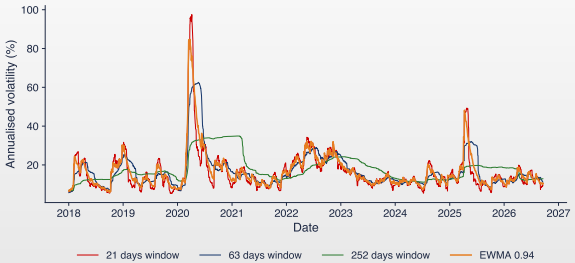
- The true variance and the proxy are both 1. Two forecasts: $h = 0.5$ (half) and $h = 2$ (double). Model: A9.
- 1. Compute the MSE of each forecast.
- 2. Compute the QLIKE of each forecast.
- 3. Say which loss punishes under-prediction more.
- Report: four numbers and one sentence on why this matters for VaR.

B1: Historical and EWMA Volatility of the S&P 500 [Solved]

- **Question**: how different are the 21-, 63- and 252-day and EWMA volatilities of the S&P 500?
- **Inputs**: S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Compute the rolling 21-, 63- and 252-day volatility and the EWMA volatility ($\lambda = 0.94$), annualised with the actual frequency.
  2. Draw the four series since 2018.
  3. Report the four values at the end of the sample and their maxima in 2020, with dates.
  4. Find the day of June 2020 when the 63-day volatility falls most, and report the change of EWMA on that day.
  5. Interpretation: which estimate would you report to a risk committee on that day?
- **Report**: the chart, twelve numbers and two sentences
- Notebook: section B1

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B1: Solution [Solved]



- $A = 252$; whole sample 18.0%; on 18 September 2026: 21 days 9.6%, 63 days 11.1%, 252 days 12.9%, EWMA 10.4%
- 2020 maxima: 97.5% (6 April 2020), 62.6% (20 May 2020), 34.7% (28 December 2020), EWMA 84.8% (24 March 2020)
- On 15 June 2020 the 63-day volatility fell from 50.4% to 43.0% (the crash day of 16 March left the window); EWMA moved from 35.6% to 34.6% (-0.9 percentage points)
- Interpretation: EWMA; the drop of the 63-day estimate is the ghost effect, not news; the 252-day value still reflects March

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B2: BET and Bitcoin [Proposed]

- **Question**: how volatile are the BET and Bitcoin compared with the S&P 500, today and at the end of 2017? Model: B1.
- **Inputs**: BET since 1997, Bitcoin since 2014, S&P 500 since 1990, daily log returns in %
- **Tasks**
  1. Compute the number of observations per year and the whole-sample annual volatility of each series.
  2. Annualise Bitcoin also with 252 and give the error in %.
  3. Report the 252-day volatility at the end of 2017 and at the end of the sample, and the maximum of the 63-day volatility with its year.
  4. Interpretation: has Bitcoin become an asset with equity-like volatility?
- **Report**: one table and two sentences
- Notebook: section B2

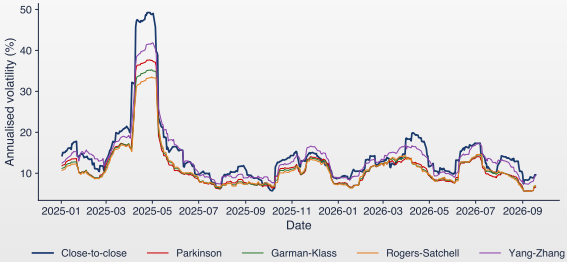
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B3: Range-Based Estimators for the S&P 500 [Solved]

- **Question**: do range-based estimators give the same volatility as closing prices for the S&P 500?
- **Inputs**: S&P 500 OHLC since 2008 ($T = 4\,707$ days)
- **Tasks**
  1. Compute o_t, u_t, d_t, c_t and the five estimators on the whole sample, annualised.
  2. Decompose $\mathrm{Var}(r_t) = \mathrm{Var}(o_t) + \mathrm{Var}(c_t) + 2\,\mathrm{Cov}(o_t, c_t)$.
  3. Draw the rolling 21-day estimates since 2025.
  4. Interpretation: why are Parkinson, Garman–Klass and Rogers–Satchell below close-to-close?
- **Report**: the chart, five volatilities, a decomposition and two sentences
- Notebook: section B3

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B3: Solution [Solved]



- Annual volatility: CC 19.9%, Parkinson 15.4%, GK 14.6%, RS 14.4%, YZ 16.3%
- Daily variances (%^{2}): $1.568 = 0.176 + 1.194 + 0.197$; the night is 13% of $\mathrm{Var}(o) + \mathrm{Var}(c)$; $\mathrm{corr}(o, c) = 0.22$
- Interpretation: these estimators see only the trading session, so they miss the overnight variance and the covariance term; YZ adds the night but not the covariance

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B4: DAX, Bitcoin and TLV [Proposed]

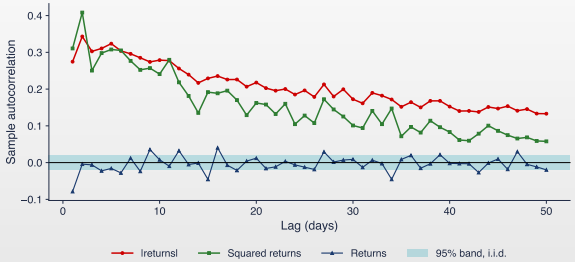
- **Question**: for which of the DAX, Bitcoin and Banca Transilvania do range-based estimators agree with close-to-close? Model: B3.
- **Inputs**: DAX OHLC since 2006, Bitcoin since 2014, TLV adjusted OHLC since 2010
- **Tasks**
  1. Compute the five annualised estimators for each asset.
  2. Compute the share of the night and $\mathrm{corr}(o_t, c_t)$.
  3. Rank the assets by the ratio Parkinson / CC.
  4. Interpretation: for which asset could you replace close-to-close by Parkinson?
- **Report**: one table and two sentences
- Notebook: section B4

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B5: Clustering in the S&P 500 [Solved]

- **Question**: are the size and the sign of S&P 500 returns equally predictable?
- **Inputs**: S&P 500 closes since 1990, daily log returns in %
- **Tasks**
  1. Draw the ACF of r_t, r_t^2 and $|r_t|$ for lags 1–50 with the i.i.d. 95% band.
  2. Report $\hat{\rho}_1, \ldots, \hat{\rho}_5$ of r_t and of r_t^2.
  3. Compute the Ljung–Box $Q(10)$ of r_t, r_t^2 and $|r_t|$, and the ARCH-LM(5) statistic with its R^2.
  4. Interpretation: does volatility clustering contradict weak-form market efficiency?
- **Report**: the chart, ten autocorrelations, four statistics and two sentences
- Notebook: section B5

B5: Solution [Solved]



- $\hat{\rho}_{\mathbf{1..5}}(r)$: -0.079, -0.004, -0.006, -0.022, -0.016; $\hat{\rho}_{\mathbf{1..5}}(r^{\mathbf{2}})$: 0.311, 0.408, 0.250, 0.298, 0.308; $T = 9\,240$
- $Q(10)$: r 90.8 ($\mathbf{p} < 0.001$), $r^{\mathbf{2}}$ 8 015, $|r|$ 8 323 (critical value 18.31); ARCH-LM(5) $= 9\,235 \times 0.241 = 2\,227$ (critical value 11.07)
- **Interpretation:** no; the size is predictable, the sign hardly; efficiency is about predictable returns, not about predictable risk (Chapter 7)

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B6: Four More Series [Proposed]

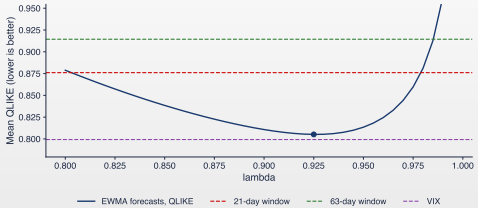
- **Question**: does volatility clustering in the BET, Bitcoin, Banca Transilvania and OMV Petrom depend on the order of the days? Model: B5.
- **Inputs**: BET since 1997, Bitcoin since 2014, TLV and SNP since 2010
- **Tasks**
  1. Compute $\hat{\rho}_1(r^2)$, $Q(10)$ of r_t^2, ARCH-LM(5) and the excess kurtosis of each series.
  2. Shuffle the returns at random (seed 2026) and compute $Q(10)$ of r_t^2 and ARCH-LM(5) again.
  3. Interpretation: why does shuffling remove the clustering but not the heavy tails?
- **Report**: one table and two sentences
- Notebook: section B6

B7: Which Forecast Is Best? [Solved]

- **Question**: which simple forecast of tomorrow's S&P 500 variance has the smallest loss since 2010?
- **Inputs**: S&P 500 returns since 1990, VIX closes; forecasts for 4 202 days from 2010
- **Tasks**
  1. Build next-day forecasts: rolling means of r_t^2 on 21, 63 and 252 days, EWMA with $\lambda = 0.94$ and 0.97, and VIX^2/A.
  2. Compute MSE and QLIKE with the proxy r_t^2.
  3. Run the Diebold–Mariano test for EWMA 0.94 against the 21-day window and for the VIX against EWMA 0.94.
  4. Draw the QLIKE of EWMA for λ from 0.80 to 0.995.
  5. Interpretation: is the VIX a better forecast of tomorrow's variance than EWMA?
- **Report**: a table of twelve numbers, two tests, the chart and two sentences
- Notebook: section B7

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B7: Solution [Solved]



	21	63	252	EWMA 0.94	EWMA 0.97	VIX
MSE	18.99	20.77	21.43	17.77	19.02	16.70
QLIKE	0.876	0.914	1.099	0.808	0.844	0.799

- DM: EWMA 0.94 against 21 days $t = -4.50$ ($p < 0.001$); VIX against EWMA 0.94 $t = -0.32$ ($p = 0.751$); best $\lambda = 0.925$
- Interpretation: no; the VIX has slightly lower losses, but the difference is not significant; both beat the rolling windows

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B8: The Best λ for the BET and Bitcoin [Proposed]

- **Question**: is the RiskMetrics $\lambda = 0.94$ a good choice for the BET and for Bitcoin? Model: B7.
- **Inputs**: BET since 1997 and Bitcoin since 2014; forecasts from 2010 (BET) and from 2016 (Bitcoin)
- **Tasks**
  1. Compute the QLIKE of next-day EWMA forecasts for λ from 0.80 to 0.995 and find the best λ.
  2. Report the QLIKE at the best λ, at 0.94 and at 0.97, and the half-life of the best λ.
  3. Run the Diebold–Mariano test of the best λ against 0.94.
  4. Interpretation: would you use $\lambda = 0.94$ for both series?
- **Report**: one table and two sentences
- Notebook: section B8

C1: Which Estimator for BVB Stocks? [Proposed]

- **Question**: do range-based estimators improve volatility forecasts for thinly traded stocks of the Bucharest Stock Exchange?
- **Inputs**: adjusted OHLC of TLV, SNP, BRD and TGN since 2010; S&P 500 since 2008 as a benchmark; models: B3, B7
- **Tasks**
  1. Compute the share of trading days with an unchanged close, and the ratios Parkinson / CC and YZ / CC of the whole-sample volatility.
  2. Compute the QLIKE (proxy r_t^2) of next-day forecasts by the 21-day CC, Parkinson and YZ variances, from 2010.
  3. Compare the four stocks with the S&P 500.
  4. List two reasons why the best estimator may differ across stocks.
  5. Interpretation: what data would you need to explain the differences?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the volatility of the S&P 500 and of Bitcoin. The answer:
- (a) Since 2008 the Parkinson volatility of the S&P 500 is 15.4% and the close-to-close volatility 19.9%; Parkinson is about five times more efficient, so the true volatility is 15.4%.
- (b) The Ljung-Box Q(10) of S&P 500 returns is 90.8 (p < 0.001), so the S&P 500 shows volatility clustering.
- (c) EWMA with lambda = 0.94 has a half-life of about 94 days.
- (d) The daily standard deviation of Bitcoin is 3.49%, so its annual volatility is 3.49 x sqrt(252) = 55.3%.
- (e) Since 1990 the VIX was above the realised volatility of the next 21 days on about 86% of days.
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of five verdicts with one line of justification each.

Points to Remember

- Annualise with the actual frequency: 252 for stock markets, 365 for Bitcoin
- Short windows react fast, long windows smooth the estimate; EWMA reacts without ghost effects
- Range-based estimators are precise but see only the session: use Yang–Zhang when the market closes at night
- Clustering is tested on r_t^2 (Ljung–Box, ARCH-LM), not on r_t
- Compare forecasts with QLIKE and test the difference; an AI answer is a draft to check

After the Seminar

- Lecture 8 develops each topic of today: historical and EWMA volatility, range-based and realised volatility, clustering, the VIX, forecast evaluation
- C1 can grow into a team project: all BET stocks, liquidity groups, Diebold–Mariano tests
- Reading: Franke, Härdle & Hafner (2019), Ch. 13; exercises in Borak, Härdle & López-Cabrera (2013), Ch. 13; Tsay (2010), Ch. 3.1

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